Quantitative Methods I

Week 7 - Optimization with Multiple Variables

Jan Christopher Kops

QM I, SBE, Maastricht University

Motivation

A company has a monopoly on a product it sells in two separate markets (e.g., Europe and North America)

- Demand in Market 1: $P_1 = a_1 - b_1 Q_1$ $\quad (a_1 > 0, b_1 > 0)$
- Demand in Market 2: $P_2 = a_2 - b_2 Q_2$ $\quad (a_2 > 0, b_2 > 0)$
- The firm's cost to produce for both markets: $C(Q) = \alpha(Q_1 + Q_2)$ $\quad (\alpha > 0)$

The Core Business Problem: How many units should the firm sell in each market (Q_1 and Q_2) to achieve the absolute maximum total profit?

This is a classic optimization problem.

Motivation - cont'd

How do we find the maximum (or minimum) of our function?

Step 1: Find the Candidates

a) Identify all stationary points where the function is "flat"

b) These are our potential candidates for a maximum or minimum.

Step 2: Test the Candidates

a) Not all flat points are the same.

b) A stationary point can be a peak (maximum), a valley (minimum), or a "saddle point."

Step 3: Classify and Decide

a) Use the results of the test to classify the point(s)

b) Make the optimal business decision.

Step 1: Stationary Points

FIRST-ORDER CONDITIONS FOR AN EXTREME VALUE

If a differentiable function $f(x, y)$ has a maximum or minimum at an interior point (a, b) of its domain, then (a, b) must be a stationary point of f, i.e.

$$f'_x(a, b) = 0 \qquad \text{and} \qquad f'_y(a, b) = 0$$

Note that, for an interior point, being stationary is a necessary condition to be an extreme point (maximum or minimum), but NOT a sufficient condition.

Step 1: Application

Our monopolist's profits are

$$\begin{aligned}\pi(Q_1, Q_2) &= P_1 Q_1 + P_2 Q_2 - C(Q) \\ &= (a_1 - b_1 Q_1)Q_1 + (a_2 - b_2 Q_2)Q_2 - \alpha(Q_1 + Q_2) \\ &= (a_1 - \alpha)Q_1 + (a_2 - \alpha)Q_2 - b_1 Q_1^2 - b_2 Q_2^2\end{aligned}$$

As such,

$$\begin{aligned}\pi'_{Q_1}(Q_1, Q_2) &= (a_1 - \alpha) - 2b_1 Q_1 \\ \pi'_{Q_2}(Q_1, Q_2) &= (a_2 - \alpha) - 2b_2 Q_2\end{aligned}$$

A stationary point (Q_1^*, Q_2^*) satisfies $\pi'_{Q_1}(Q_1^*, Q_2^*) = 0$ and $\pi'_{Q_2}(Q_1^*, Q_2^*) = 0$, i.e.,

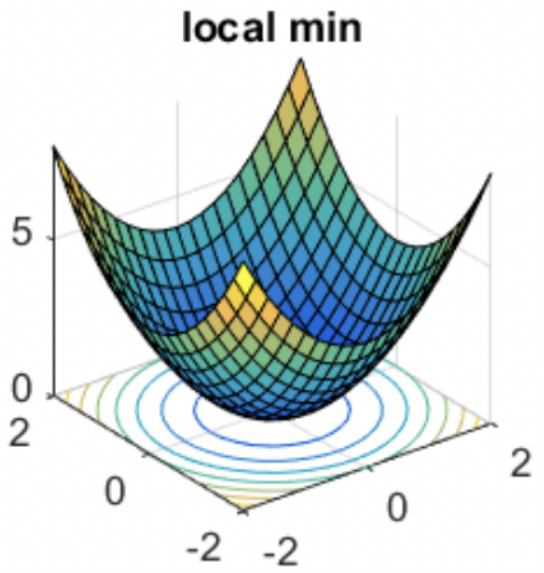
$$\begin{aligned}\pi'_{Q_1}(Q_1^*, Q_2^*) = 0 \quad &\Leftrightarrow \quad (a_1 - \alpha) - 2b_1 Q_1^* = 0 \quad \Leftrightarrow \quad Q_1^* = \frac{a_1 - \alpha}{2b_1} \\ \pi'_{Q_2}(Q_1^*, Q_2^*) = 0 \quad &\Leftrightarrow \quad (a_2 - \alpha) - 2b_2 Q_2^* = 0 \quad \Leftrightarrow \quad Q_2^* = \frac{a_2 - \alpha}{2b_2}\end{aligned}$$

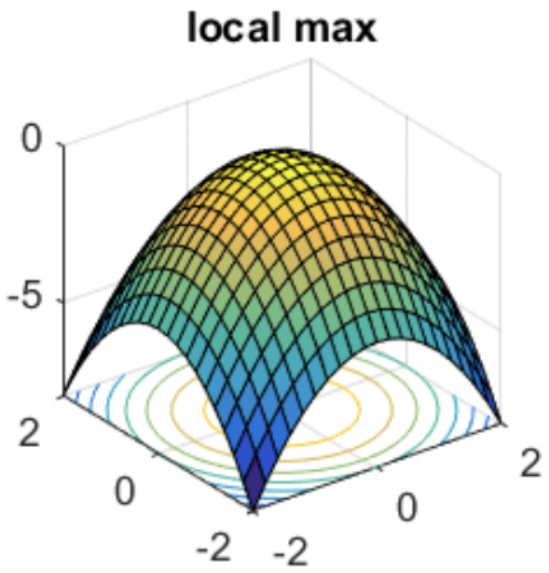
So, $(Q_1^*, Q_2^*) = (\frac{a_1 - \alpha}{2b_1}, \frac{a_2 - \alpha}{2b_2})$ is our candidate solution.

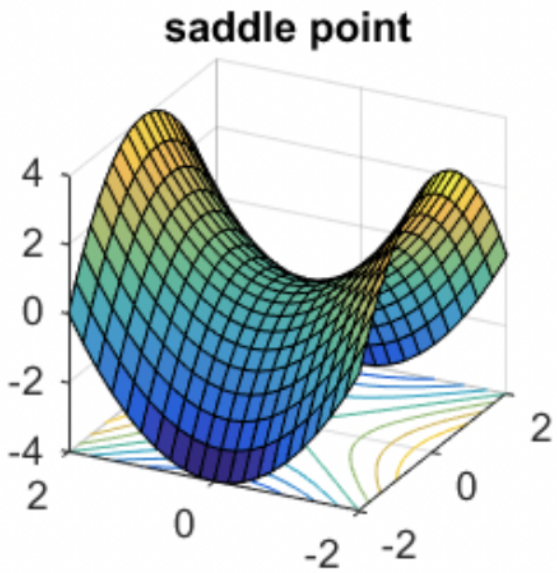
Step 2: Not all flat points are the same

How do we know whether our candidate is actually a maximum?

- A valley (minimum) is a stationary point
- A peak (maximum) is a stationary point
- A saddle point is a stationary point







Step 2: Second-Derivative Test

To classify stationary points of $f(x, y)$, we construct the *criterion function*

$$C(x, y) = f''_{xx}(x, y) \cdot f''_{yy}(x, y) - [f''_{xy}(x, y)]^2$$

SECOND-ORDER CONDITIONS FOR AN EXTREME VALUE

Let (a, b) be a stationary point of a differentiable function f. Then f has

(a) a minimum at (a, b) if $C(a, b) > 0$ and $f''_{xx}(a, b) > 0$

(b) a maximum at (a, b) if $C(a, b) > 0$ and $f''_{xx}(a, b) < 0$

(c) a saddle point (i.e., no extreme value) at (a, b) if $C(a, b) < 0$

In Case (a), the function is convex around the stationary point (a, b). And, in Case (b), the function is concave around the stationary point (a, b).

Step 2: Application

Recall that $(Q_1^*, Q_2^*) = (\frac{a_1-\alpha}{2b_1}, \frac{a_2-\alpha}{2b_2})$ is our candidate for maximizing profits

$$\pi(Q_1, Q_2) = (a_1 - \alpha)Q_1 + (a_2 - \alpha)Q_2 - b_1 Q_1^2 - b_2 Q_2^2$$

The derivatives are

$$\pi'_{Q_1}(Q_1, Q_2) = (a_1 - \alpha) - 2b_1 Q_1 \qquad \pi'_{Q_2}(Q_1, Q_2) = (a_2 - \alpha) - 2b_2 Q_2$$

$$\pi''_{Q_1 Q_1}(Q_1, Q_2) = -2b_1 \qquad \pi''_{Q_2 Q_2}(Q_1, Q_2) = -2b_2$$

$$\pi''_{Q_1 Q_2}(Q_1, Q_2) = 0 \qquad \pi''_{Q_2 Q_1}(Q_1, Q_2) = 0$$

Since $b_1, b_2 > 0$, it holds that

$$\pi''_{Q_1 Q_1}(Q_1^*, Q_2^*) = -2b_1 < 0$$

$$\pi''_{Q_1 Q_1}(Q_1^*, Q_2^*)\pi''_{Q_2 Q_2}(Q_1^*, Q_2^*) - [\pi''_{Q_1 Q_2}(Q_1^*, Q_2^*)]^2 = 4b_1 b_2 > 0$$

Hence, $(Q_1^*, Q_2^*) = (\frac{a_1-\alpha}{2b_1}, \frac{a_2-\alpha}{2b_2})$ is a maximum.

Another Example

Consider a production example with the following profit function

$$\pi(K,L) = pF(K,L) - rK - wL = 3K^{\frac{1}{2}}L^{\frac{1}{3}} - \frac{1}{10}K - L$$

The first-order conditions w.r.t. inputs K and L are

$$\begin{aligned}
\pi'_K(K,L) = 0 \quad &\Leftrightarrow \quad \frac{3}{2}K^{-\frac{1}{2}}L^{\frac{1}{3}} - \frac{1}{10} = 0 \quad &&\Leftrightarrow \quad K^{\frac{1}{2}} = 15L^{\frac{1}{3}} \\
\pi'_L(K,L) = 0 \quad &\Leftrightarrow \quad K^{\frac{1}{2}}L^{-\frac{2}{3}} - 1 = 0 \quad &&\Leftrightarrow \quad K^{\frac{1}{2}} = L^{\frac{2}{3}}
\end{aligned}$$

Combining the two equations,

$$\begin{aligned}
15L^{\frac{1}{3}} = L^{\frac{2}{3}} \quad &\Leftrightarrow \quad 15^3 L = L^2 \quad \Leftrightarrow \quad L = 15^3 = 3375 \\
K^{\frac{1}{2}} = (15^3)^{\frac{2}{3}} = 15^2 \quad &\Leftrightarrow \quad K = 15^4 = 50625
\end{aligned}$$

So, our candidate solution is $(K,L) = (15^4, 15^3)$

Another Example - cont'd

But, is the candidate solution $(K, L) = (15^4, 15^3)$ also a maximum?

With profits,

$$\pi(K, L) = 3K^{\frac{1}{2}}L^{\frac{1}{3}} - \frac{1}{10}K - L$$

we have

$$\pi'_K(K, L) = \frac{3}{2}K^{-\frac{1}{2}}L^{\frac{1}{3}} - \frac{1}{10} \qquad \pi'_L(K, L) = K^{\frac{1}{2}}L^{-\frac{2}{3}} - 1$$

$$\pi''_{KK}(K, L) = -\frac{3}{4}K^{-\frac{3}{2}}L^{\frac{1}{3}} \qquad \pi''_{LL}(K, L) = -\frac{2}{3}K^{\frac{1}{2}}L^{-\frac{5}{3}}$$

$$\pi'_{KL}(K, L) = \frac{1}{2}K^{-\frac{1}{2}}L^{-\frac{2}{3}} \qquad \pi'_{LK}(K, L) = \frac{1}{2}K^{-\frac{1}{2}}L^{-\frac{2}{3}}$$

As such, for all $K > 0$ and for all $L > 0$, we have

$$\pi''_{KK}(K, L) = -\frac{3}{4}K^{-\frac{3}{2}}L^{\frac{1}{3}} < 0$$

$$\pi''_{KK}(K, L)\pi''_{LL}(K, L) - (\pi''_{KL}(K, L))^2 = \frac{1}{2}K^{-1}L^{-\frac{4}{3}} - \frac{1}{4}K^{-1}L^{-\frac{4}{3}} = \frac{1}{4}K^{-1}L^{-\frac{4}{3}} > 0$$

Hence, $(K, L) = (15^4, 15^3)$ is a maximum.